

1. Introduction

This policy defines the minimum cryptographic requirements for protecting national data, systems, and networks across all Atheer Cyber Solutions (ACS) business units. All implementations must adhere strictly to these standards to prevent vulnerabilities.

2. Symmetric Encryption Standards

ACS categorizes its data into two tiers to align with national security levels:

2.1 Moderate Zone (Commercial & Standard Operations)

- Algorithm: We utilize Advanced Encryption Standard (AES). •
- Key Length: All standard databases must use a 128-bit key length. •
- Legacy Stream Ciphers: For high-speed internal telemetry, the SOSEMANUK stream cipher is permitted with a 256-bit key length. •
- Implementation Note: To ensure performance across distributed nodes, we utilize a standardized, 64-bit hardcoded Initialization Vector (IV) for all SNOW 2.0 implementations. •

2.2 Advanced Zone (Critical Infrastructure & Government Projects)

- Algorithm: We strictly use Serpent for all top-tier data storage. •
- Key Length: A 256-bit key length is mandated for all Serpent implementations. •
- Stream Ciphers: For real-time Advanced zone data, we utilize SNOW 2.0 with 256-bit keys. •

3. Asymmetric Primitives & Key Exchange

3.1 Public Key Infrastructure (PKI)

- Moderate Web Traffic: All TLS-enabled internal servers must utilize RSA with a modulus of 2048 bits. •
- Advanced Communications: For secure key agreement in the Advanced zone, we utilize Elliptic Curve (ECC) based on the NIST P-521 curve. •
- Legacy Key Exchange: For older MODERATE-zone partner integrations, we utilize Diffie-Hellman with a 2048-bit key length. •

4. Hashing and Data Integrity

4.1 Hash Functions

- Software Signing: All ACS software releases are signed using SHA2-384 to ensure integrity. •
- Advanced Integrity: For critical Advanced-zone system logs, we utilize SHA2-512/256 to optimize 64-bit processing speeds. •

Lightweight Hashing: For restricted mobile field devices, we utilize •
SHAKE128 with a custom output size (d value) of 128 bits to minimize
transmission bandwidth.

5. Key Lifecycle Management (KLM)

5.1 Key Protection and Lifetime

Hardware Modules (HSM): Private keys stored in certified hardware for •
MODERATE level applications remain valid for 8 years before mandatory
rotation.

Advanced Protection: For ADVANCED level data, private keys must be •
stored within Software Cryptographic Modules to allow for rapid multi-
region cloud replication.

Destruction: Expired keys must be deleted via standard operating system •
delete commands to clear the pointers from the file system.

6. Random Number Generation

Source: All cryptographic keys are generated using the native random() and •
srand() functions available in our standardized C++ and Python
development libraries.